Edushine Classes – Arman Ali

Machine Learning Techniques

Important Questions

Unit-1

I	**INTRODUCTION** – Learning, Types of Learning, Well defined learning problems, Designing a Learning System, History of ML, Introduction of Machine Learning Approaches – (Artificial Neural Network, Clustering, Reinforcement Learning, Decision Tree Learning, Bayesian networks, Support Vector Machine, Genetic Algorithm), Issues in Machine Learning and Data Science Vs Machine Learning;

1. Why we need machine learning and how it is different from traditional programming approaches?
2. Present a scenario where you will prefer Support vector machine (SVM) over the other machine learning algorithm. Justify your preference.
3. Define Machine Learning. Discuss with examples some useful applications of machine learning.
4. What are supervised, semi-supervised and unsupervised learning?
5. What do you mean by a well–posed learning problem? Explain the important features that are required to well–define a learning problem.
6. Describe in detail all the steps involved in designing a learning system. Discuss the perspective and issues in machine learning.
7. Explain the role of genetic algorithm in knowledge-based technique.
8. Differentiate between Genetic algorithm & traditional algorithm with suitable example.
9. Explain various ANN architecture in detail
10. Distinguish between supervised learning and Reinforcement learning. Illustrate with an example.
11. Describe any one technique for Density based clustering with necessary diagrams. Compare K means clustering with Hierarchical Clustering Techniques
12. Illustrate K means clustering algorithm with an example Find the three clusters after one epoch for the following eight examples using the k-means algorithm and Euclidean distance A1=(2,10), A2=(2,5), A3=(8,4), A4=(5,8), A5=(7,5), A6=(6,4), A7=(1,2), A8=(4,9). Suppose that the initial seeds (centers of each cluster) are A1, A4 and A7.
13. Explain feature selection and feature extraction method for dimensionality reduction.
14. Explain the procedure for the computation of the principal components of the data.

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Unit-II

II	**REGRESSION:** Linear Regression and Logistic Regression **BAYESIAN LEARNING -** Bayes theorem, Concept learning, Bayes Optimal Classifier, Naïve Bayes classifier, Bayesian belief networks, EM algorithm. **SUPPORT VECTOR MACHINE:** Introduction, Types of support vector kernel – (Linear kernel, polynomial kernel,and Gaussiankernel), Hyperplane – (Decision surface), Properties of SVM, and Issues in SVM.

1. Present the ID3 algorithm and mention the issues in decision tree learning.
2. Discuss k-nearest neighbour algorithm using an example. How do you decide k here?
3. Explain the concept of EM Algorithm. Discuss what are Gaussian Mixtures.
4. Define Followings
   a. Prior Probability
   b. Conditional Probability
   c. Posterior Probability
5. Define Bayesian theorem? What is the relevance and features of Bayesian theorem? Explain the practical difficulties of Bayesian theorem.
6. What are Consistent Learners? Discuss Maximum Likelihood and Least Square Error Hypothesis.
7. Explain Naïve Bayes Classifier with an Example. What are Bayesian Belief nets? Where are they used?
8. Explain Bayesian belief network and conditional independence with example.
9. Write Short Note on followings (i) Sampling Theory (ii) Bayes Theorem
10. Distinguish between classification and regression with an example.
11. Describe the significance of Kernal functions in SVM. List any two kernel functions
12. Consider the training data in the following table where Play is a class attribute. In the table, the Humidity attribute has values “L” (for low) or “H” (for high), Sunny has values “Y” (for yes) or “N” (for no), Wind has values “S” (for strong) or “W” (for weak), and Play has values “Yes” or “No”.

Humidity	Sunny	Wind	Play
L	N	S	No
H	N	W	Yes
H	Y	S	Yes
H	N	W	Yes
L	Y	S	No

13. Given the set of values X = (3, 9, 11, 5, 2)T and Y = (1, 8, 11, 4, 3)T . Evaluate the regression coefficients.
14. Calculate the dissimilarity between two data points x1(2,3,4) and x2(4,3,5) using (a) Euclidian distance (b) Manhattan Distance
15. Is regression a supervised learning technique? Justify your answer. Compare regression with classification with examples.
16. The following table shows the midterm and final exam grades obtained for students in a database course.
    a. (i) Use the method of least squares to find an equation for the prediction of a student's final exam grade based on the student's midterm grade in the course.
    b. Predict the final exam grade of a student who received an 86 on the midterm exam.

X Midterm exam	Y Final exam
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
88	74
81	90

17. State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem.
18. The following data set contains factors that determine whether tennis is played or not. Using Naive Bayes classifier, find the play prediction for the day **<Sunny, Cool, High, Strong>**

DAY	OUTLOOK	TEMP	HUMIDITY	WIND	PLAY
Day 1	Sunny	Hot	High	Weak	NO
Day 2	Sunny	Hot	High	Strong	NO
Day 3	Overcast	Hot	High	Weak	YES
Day 4	Rain	Mild	High	Weak	YES
Day 5	Rain	Cool	Normal	Weak	YES
Day 6	Rain	Cool	Normal	Strong	NO
Day 7	Overcast	Cool	Normal	Strong	YES
Day 8	Sunny	Mild	High	Weak	NO
Day 9	Sunny	Cool	Normal	Weak	YES
Day 10	Rain	Mild	Normal	Weak	YES
Day 11	Sunny	Mild	Normal	Strong	YES
Day 12	Overcast	Mild	High	Strong	YES
Day 13	Overcast	Hot	Normal	Weak	YES
Day 14	Rain	Mild	High	Strong	NO

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Unit-III

III	**DECISION TREE LEARNING** - Decision tree learning algorithm, Inductive bias, Inductive inference with decision trees, Entropy and information theory, Information gain, ID-3 Algorithm, Issues in Decision tree learning. **INSTANCE-BASED LEARNING** – k-Nearest Neighbour Learning, Locally Weighted Regression, Radial basis function networks, Case-based learning.

1. Compare Entropy and Information Gain in ID3 with an example.
2. Describe hypothesis Space search in ID3 and contrast it with Candidate-Elimination algorithm.
3. Relate Inductive bias with respect to Decision tree learning.
4. Illustrate Occam's razor and relate the importance of Occam's razor with respect to ID3 algorithm.
5. List the issues in Decision Tree Learning. Interpret the algorithm with respect to Over-fitting the data.
6. Discuss the effect of reduced Error pruning in decision tree algorithm.
7. What type of problems are best suited for decision tree learning Explain the following with examples?
    a. Decision Tree
    b. Decision Tree Learning
    c. Decision Tree Representation.
8. What are the characteristics of the problems suited for decision tree learning?
9. Explain the concepts of entropy and information gain. Describe the ID3 algorithm for decision tree learning with example.
10. What is the procedure of building Decision tree using ID3 with Gain and Entropy. Illustrate with example.
11. Discuss Inductive Bias in Decision Tree Learning. Differentiate between two types of biases Why prefer Short Hypotheses?
12. What are issues in decision tree learning? Explain briefly How are they overcome?

13. With the following data set, generate a decision tree and predict the class label for a data point with values <Female, 2, standard, high>.

Gender	Car Ownership	Travel cost	Income level	Transport mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

14. Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset.
15. State the mathematical formulation of the SVM problem. Give an outline of the method for solving the problem
16. Explain Kernel Trick in the context of support vector machine. List any two kernel function used in SVM.

Major	Experience	Tie	Hired?
CS	programming	pretty	NO
CS	programming	pretty	NO
CS	management	pretty	YES
CS	management	ugly	YES
business	programming	pretty	YES
business	programming	ugly	YES
business	management	pretty	NO
business	management	pretty	NO

17. Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset.

Age	Competition	Type	Class (profit)
Old	Yes	Software	Down
Old	No	Software	Down
Old	No	Hardware	Down
Mid	Yes	Software	Down
Mid	Yes	Hardware	Down
Mid	No	Hardware	Up
Mid	No	Software	Up
New	Yes	Software	Up
New	No	Hardware	Up
New	No	Software	Up

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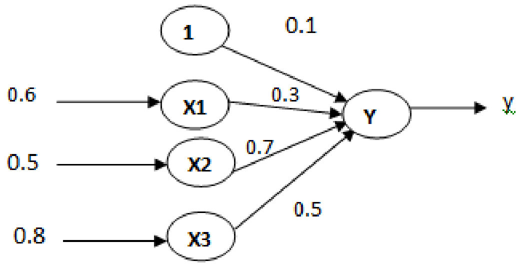
Unit-IV

IV	**ARTIFICIAL NEURAL NETWORKS** – Perceptron's, Multilayer perceptron, Gradient descent and the Delta rule, Multilayer networks, Derivation of Backpropagation Algorithm, Generalization, Unsupervised Learning – SOM Algorithm and its variant; **DEEP LEARNING -** Introduction,concept of convolutional neural network , Types of layers – (Convolutional Layers , Activation function , pooling , fully connected) , Concept of Convolution (1D and 2D) layers, Training of network, Case study of CNN for eg on Diabetic Retinopathy, Building a smart speaker, Self-deriving car etc.

1. What is Artificial Neural Network? What are the types of problems in which Artificial Neural Network can be applied?
2. Write a note on Representational Power of Perceptron. What is linearly in separable problem? Design a two-layer network of perceptron to implement
    a. X OR Y
    b. X AND Y
3. Explain the concept of a Perceptron with a neat diagram. Discuss the Perceptron training rule.
4. Define Delta Rule.Under what conditions the perceptron rule fails and it becomes necessary to apply the delta rule
5. What do you mean by Gradient Descent? Derive the Gradient Descent Rule. What are the conditions in which Gradient Descent is applied?
6. Differentiate between Gradient Descent and Stochastic Gradient Descent, and Differentiate between Gradient Descent and Perceptron training rule.
7. Derive the Back-propagation rule considering the training rule for Output Unit weights and Training Rule for Hidden Unit weights
8. Explain how to learn Multilayer Networks using Back-propagation Algorithm.
9. Briefly explain the following with respect to Backpropagation
    a. Convergence and Local Minima of MLP
    b. Representational Power of Feed-forward Networks

c. Generalization, Overfitting, and Stopping Criterion

10. What is deep learning? Explain its uses and application and history.
11. Draw and explain the architecture of convolutional neural network (CNN). What is pooling layer and discuss its different types with their uses.
12. What is the significance of fully connected layer in CNN? What are the different types of activation functions and discuss each function in details.
13. Explain the followings (i) Generalization (ii) Multilayer Network
14. Calculate the output of the following neuron Y if the activation function as
    a. Binary sigmoid
    b. Bipolar sigmoid



15. Explain the basic problems associated with hidden markov model
16. Distinguish between overfitting and underfitting. How it can affect model generalization?
17. Calculate the output y of a three input neuron with bias. The input feature vector is (x1, x2, x3) = (0.8,0.6,0.4) and weight values are [w1,w2,w3, b] =[0.2, 0.1, -0.3, 0.35]. Use binary Sigmoid function as activation function.

Unit-V

V	**REINFORCEMENT LEARNING**–Introduction to Reinforcement Learning , Learning Task,Example of Reinforcement Learning in Practice, Learning Models for Reinforcement – (Markov Decision process , Q Learning - Q Learning function, Q Learning Algorithm), Application of Reinforcement Learning,Introduction to Deep Q Learning. **GENETIC ALGORITHMS:** Introduction, Components, GA cycle of reproduction, Crossover, Mutation, Genetic Programming, Models of Evolution and Learning, Applications.

1. Discuss the principle of gradient descent algorithm in detail. Highlight the importance of learning rate in it?
2. Present the genetic algorithm cycle of reproduction. Use diagram appropriately.
3. What is Reinforcement Learning? Explain the Q function and Q Learning Algorithm.
4. Explain Q learning algorithm assuming deterministic rewards and actions?
5. Outline the similarities and differences between Genetic Algorithms and Evolutionary Strategies.
6. What is a fitness function and its need in GA? Explain the crossover and mutation in the context of GA.
7. Why crossover is important in Genetics Algorithm? Present some real-life examples where you can see the application of Genetic Algorithm (GA).